

## 10. Analysis A: noise vs Z

The noise is split into three parts: temporal (frame-to-frame), fixed-pattern (static, tied to the projector pattern and pixel position), and total. The temporal part is fitted to a disparity-noise model, which yields the forward-model parameters.

1. Load and mask. Convert depth codes to mm with DEPTH\_LSB\_MM and set no-reads to NaN. Project the plate outline into the image using the registered pose. Shrink it by BOUNDARY\_BAND\_HALF\_WIDTH\_PX to form the region of interest (ROI).
2. Ground truth. For each ROI pixel, intersect its ray with the registered plate plane. This gives  $Z_{GT}(u, v)$ .
3. Temporal noise. For each pixel, compute  $\sigma_t(u, v)$  as the standard deviation over frames (ddof = 1). Report the ROI median and 95th percentile for each station. Save the  $\sigma_t$  map, since its structure shows the projector-dot layout.
4. Fixed-pattern noise and bias. Average the frames to get  $\bar{Z}(u, v)$ . Fit a free plane to  $\bar{Z}$ ; the standard deviation of the residual is  $\sigma_{fp}$ . The mean of  $(\bar{Z} - Z_{GT})$  over the ROI is  $\text{bias}(Z)$ . Also report the angle between the fitted plane and the registered plane.
5. Total noise. For each frame, compute the residual from  $Z_{GT}$ .  $\sigma_{tot}$  is the RMS over the ROI and frames. Check that  $\sigma_{tot}^2 \approx \sigma_t^2 + \sigma_{fp}^2 + \text{bias}^2$ ; a large mismatch points to an analysis error or to non-stationary noise.
6. Fill rate. Report the fraction of valid pixels in the ROI, for each station and tilt.
7. Spatial correlation. Compute the normalized 2-D autocorrelation of the temporal component ( $Z - \bar{Z}$ ), averaged over frames, and of the fixed-pattern residual map. The correlation length along H and along V is the lag where the normalized autocorrelation first falls to 1/e. Compare it with the Tier-A IDW neighborhood N and with PHASE\_JITTER\_SPAN\_PX; the span must exceed it.
8. Quantization. Histogram the raw depth codes for the central  $20 \times 20$  px (QUANTIZATION\_PATCH\_PX) over all frames. The most common spacing between populated levels is  $\delta Z_q(Z)$ . If  $\delta Z_q$  equals the depth LSB (DEPTH\_LSB\_MM) at every Z, the output LSB is the quantizer and disparity quantization cannot be seen here; use the B-Z ramp instead. Otherwise compute  $q = \delta Z_q \cdot k / Z^2$  at each Z. It should be constant if the quantizer works in disparity.
9. Noise model. Fit  $\sigma_t(Z)$  by weighted least squares. The first term,  $\sigma_0$ , is a floor that does not depend on Z (LSB/ $\sqrt{12}$  or electronics). The second is disparity noise  $\sigma_d$  propagated through  $d =$

$k/Z$ . Also fit a free power law  $\sigma_t = a \cdot Z^n$ . Triangulation predicts  $n \approx 2$ , and a clearly different  $n$  is a finding worth reporting.

$$\sigma_t(Z) = \sqrt{\sigma_0^2 + \left(\frac{\sigma_d Z^2}{k}\right)^2}$$

1. Incidence angle. From the tilt sub-series, plot  $\sigma_t$ ,  $\sigma_{tot}$ , and fill rate against incidence angle at each  $Z$ .
2. Drift correction. Plot the sentinel mean  $Z$  against time and sensor temperature, each mounted target relative to its first sentinel after mounting (Section 5, Step 3). If the drift exceeds  $WARMUP\_DRIFT\_FRACTION\_OF\_SIGMA \times \sigma_t$ , subtract a time-interpolated sentinel offset from the bias estimates and report the size of the correction.
3. Continuity with earlier data. Also compute the existing `testZRepeatabilityBrownBoard.py` metrics (boxes of half side `LEGACY_BOX_HALF_PX`: mean min–max range and mean per-pixel std) at the `LEGACY_METRIC_DEPTHS_MM` stations, which the A series adds to the ladder for this purpose. This lets the new results be compared directly with the existing VSX3000 data sets.
4. Outputs. `A_noise_summary.csv` with one row per station. Figures (PNG and SVG): the three  $\sigma$  curves vs  $Z$  with the model fit; fill rate vs  $Z$ ;  $\sigma_t$  maps; autocorrelation profiles; depth-code histograms. The fitted  $\sigma_d$ ,  $\sigma_0$ ,  $q$ ,  $k$ , and correlation length go in `forward_model_parameters.json` for the Tier-A simulator.